

EEET202

CHINES & TRANSFORMERS

MODULE:2

ure:3 – Armature Reaction

e performance characteristics of DC generators

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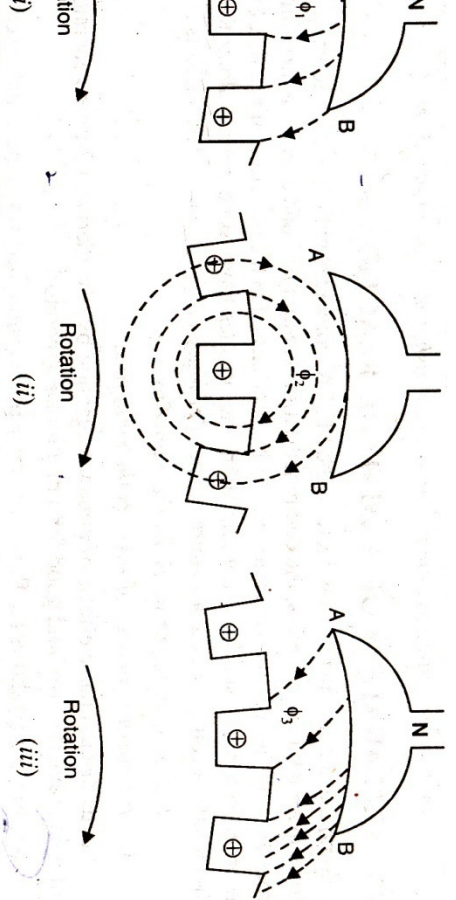
ARMATURE REACTION

In a dc machine which is developed by main field winding, the main flux is developed by the main field winding.

When a dc machine is running, it also carries armature current, it also carries armature flux and it is called armature flux.

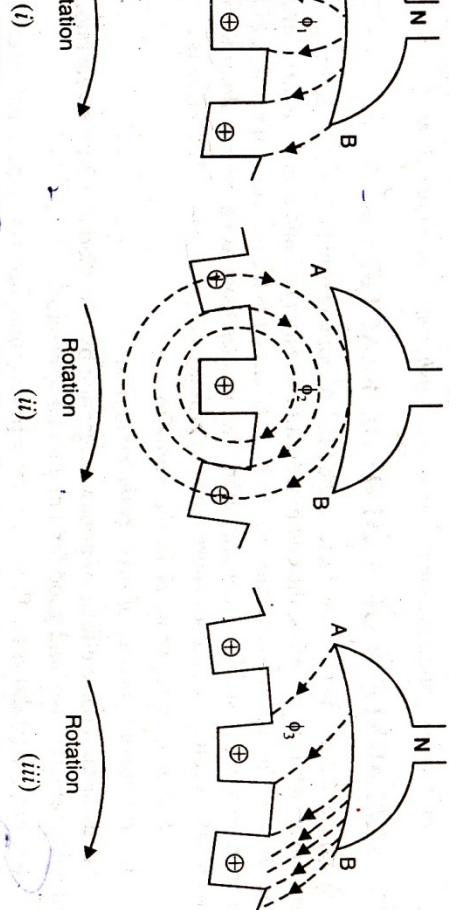
The armature flux **distorts and weakens** the main flux.

The effect of armature flux on main flux is known as armature reaction.



ARMATURE REACTION (contd...)

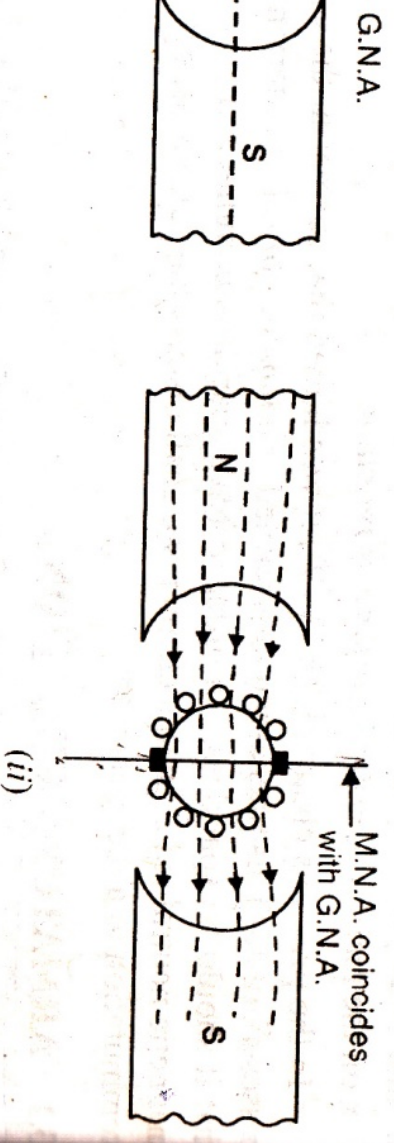
of armature reaction in DC machine
on the main flux (Cross-magnetisation)
on the main flux (demagnetisation)

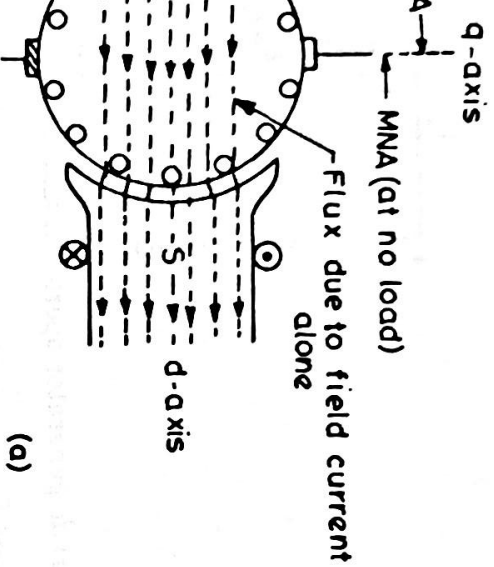


Geometrical & Magnetic Neutral Axes

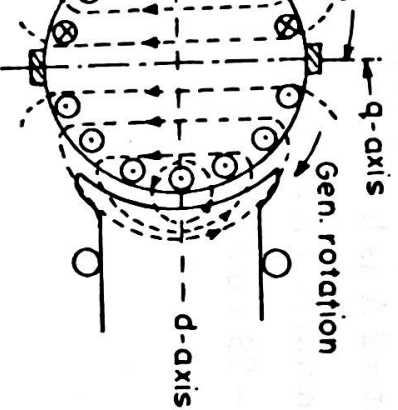
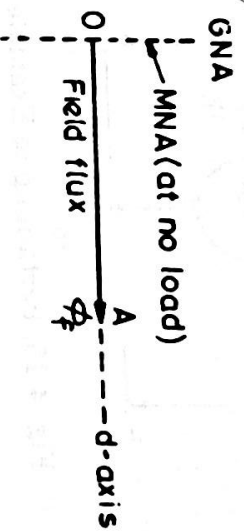
Geometrical Neutral Axis (G.N.A.): The axis that bisects the space between the centre line of adjacent poles.

Magnetic Neutral Axis (M.N.A.): The axis drawn perpendicular to the mean direction of the flux passing through the centre of the armature. No emf is produced in conductors along this axis, because they are perpendicular to the flux. So, in order to achieve sparkless commutation, the brushes must lie along M.N.A.

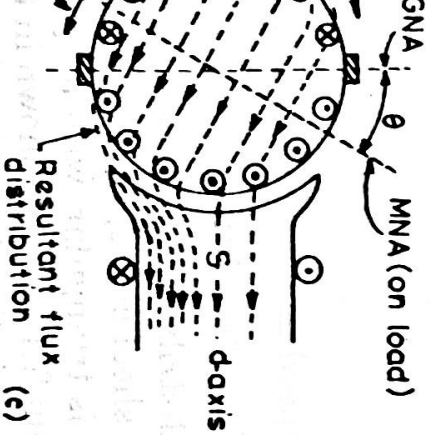
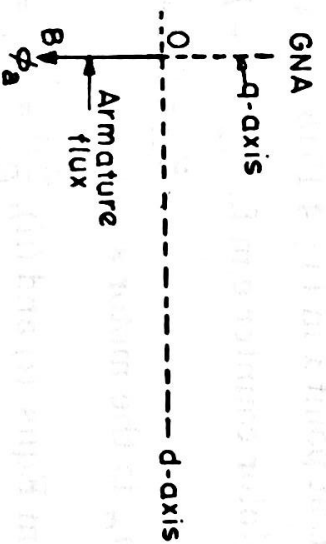




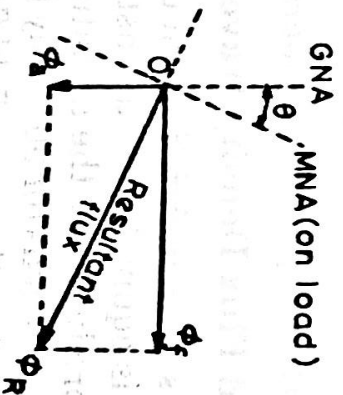
(a)



(b)



(c)



Effects of Armature Reaction

At in armature conductors, the flux across the main and MNA coincides with GNA.

Flows in armature conductors, the combined flux and armature flux (armature reaction) shifts by an angle θ from GNA.

Therefore, MNA shifts in the direction of rotation of the motor, to have sparkless commutation, the brushes have to be shifted by the angle θ from the new MNA.

On shift, armature mmf is also shifted by the same angle θ . At armature mmf ϕ_a will no longer be vertically aligned with the direction of rotation of the motor, angle θ .

Effect of Brush Shift

resolved in to 2 rectangular components ϕ_c and ϕ_a . ϕ_c is in direct opposition to the main mmf. So it has a weakening effect on main flux and it is called **(weakening) component of armature reaction**.

ϕ_a is perpendicular to the main mmf. So it is in cross flux. Hence it is called **cross-magnetising component of armature reaction**.

Demagnetising effect leads to reduced generated voltage. Cross-magnetising effect leads to sparking at the brushes.

